Session 5 (DATA): Exercise sheet

Foundations of Clinical Research · Level 1 · Neudata · *#ClearDataClearImpact* · Week 5 · Saturday 28 November 2026 · online, 19:30–22:30

Name: _____ **Department:** _____

Type your answers in the Word file, or print the PDF. Exercises marked **[FLUID-ICU]** use the course dataset and its five results packs (`Results_Pack_Q1.pdf` to `Results_Pack_Q5.pdf`, sent by email). FLUID-ICU is simulated.

#	Exercise	When
1	Classify the variables	At home
2	Fix the bad CRF items	In class, section 5.1
3	Choose the right summary	At home
4	Judge a bad figure	At home
5	[FLUID-ICU] Spot the problems in the raw data	At home
6	[FLUID-ICU] Read the FLUID-ICU Table 1	In class, section 5.4
7	[FLUID-ICU] A FLUID-ICU question as a PECO sentence	In class, section 5.2
8	Proposal builder 5: data for your own proposal	In class (8.1), then at home

Abbreviations: CRF = case report form, ICU = intensive care unit, IQR = interquartile range, SD = standard deviation.

Exercise 1: Classify the variables

Task. For each variable, tick **one** type. Then answer the bonus question.

#	Variable (ICU study of septic shock)	Continuous	Count	Binary	Nominal	Ordinal	Time-to-event
a	Age at ICU admission (years)	☐	☐	☐	☐	☐	☐
b	Sex (male / female)	☐	☐	☐	☐	☐	☐
c	Glasgow Coma Scale on admission (3–15)	☐	☐	☐	☐	☐	☐
d	Main source of sepsis (lung / urine / abdomen / skin / other)	☐	☐	☐	☐	☐	☐
e	Number of vasopressors running at 24 h (0, 1, 2, 3)	☐	☐	☐	☐	☐	☐
f	Cumulative fluid balance at day 5 (ml/kg)	☐	☐	☐	☐	☐	☐
g	Alive at day 28 (yes / no)	☐	☐	☐	☐	☐	☐
h	Time from ICU admission to death (patients alive at day 28 are censored)	☐	☐	☐	☐	☐	☐
i	Pain: none / mild / moderate / severe	☐	☐	☐	☐	☐	☐
j	Blood group (A, B, AB, O)	☐	☐	☐	☐	☐	☐

Bonus: which ONE of these would you most like to collect as an exact number rather than in groups, and why?

Exercise 2: Fix the bad CRF items

Task. Two nurses could fill in each item below in two different ways. For each item, write what is wrong, then rewrite it so that it has **only one possible reading**. Be ready to read one rewrite aloud when the instructor asks.

Check each rewrite: Is the unit in the question? Is the time point in the question? Is only one answer possible? Is there an "unknown / not done" option?

#	Bad item	What is wrong?	Your rewritten item
a	Fluid given: _____		
b	Weight: _____		
c	Comorbidities: _____		
d	Lactate high? Y / N		

Exercise 3: Choose the right summary

Task. For each variable, write its shape or type, the summary you would report in Table 1, and the best graph.

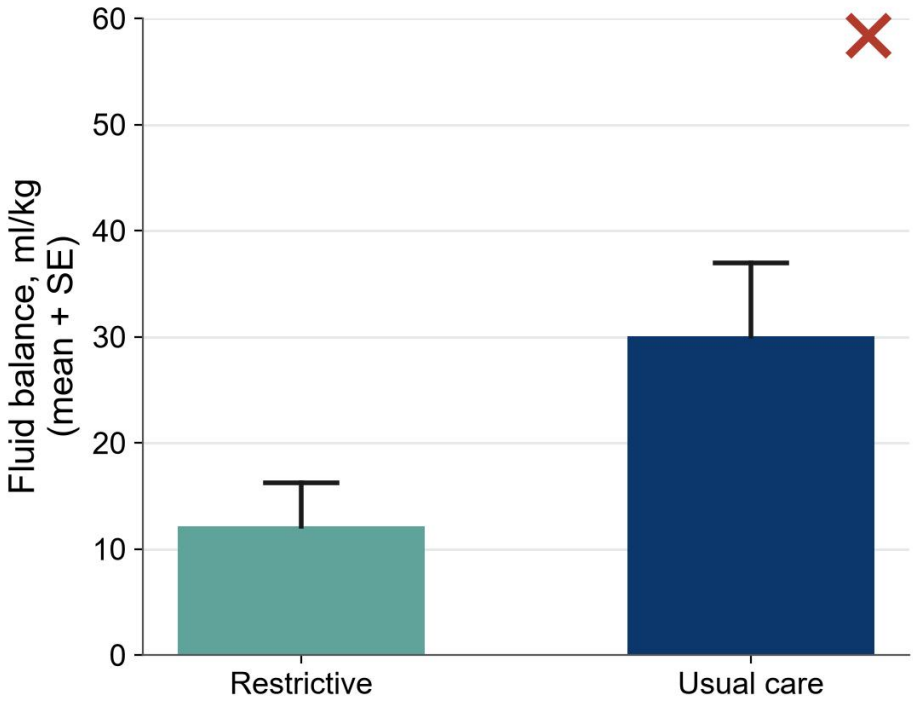
> **Choose from:** shape or type: symmetric · right-skewed · binary · nominal. Summary: mean (SD) · median (IQR) · n (%). Graph: bar chart · histogram · boxplot · scatter plot

#	Variable (200 ICU patients)	Shape / type	Summary	Graph
a	Age (years), bell-shaped			
b	ICU length of stay (days): most 1–5 days, a few > 30 days			
c	Died in hospital (yes / no)			
d	C-reactive protein (mg/L): long right tail			
e	Source of sepsis (5 categories)			
f	Systolic blood pressure, compared between two treatment groups			

Exercise 4: Judge a bad figure

Task. Look at the three posters described below and answer questions 1 to 5.

A poster in the hospital corridor shows this figure (fluid balance at day 5, 24 patients per group):



1. What does each bar show, and what does the error bar show? What is missing?
2. Name **two things the reader cannot see** in this figure.
3. Sketch how you would show the same data in a paper. What do you gain?

A second poster shows ward mortality as bars of 31% (Ward A) and 29% (Ward B), with the y-axis running from 28% to 32%.

4. What impression does it give, and how would you fix it?

A third poster puts hand-hygiene compliance (%) and infection rate on one graph with two different y-axes and writes "hygiene works!".

5. Give two reasons to be careful.

Exercise 5: [FLUID-ICU] Spot the problems in the raw FLUID-ICU data

Task. The six values below come from the raw file `fluid_icu_raw.csv`, as the four ICUs sent it. For each one, write what is wrong and what the cleaning script should do. Then answer the bonus question.

#	Column	Value in the raw file	What is wrong?	What should the script do?
a	sex	`m`, `male`, `Female`		
b	admission date	`14/03/2023` in one row, `14-Mar-2023` in another		
c	lactate (ICU C)	`23.4`		
d	lactate	`999`		
e	age	`210`		
f	patient ID	`FLU-139` appears in two identical rows		

Bonus: why must the raw file never be edited by hand? Name two things the cleaning log records for every change.

Exercise 6: [FLUID-ICU] Read the FLUID-ICU Table 1

Task. Open Table 1 (baseline characteristics by fluid-balance group) in the Q1 results pack. The Q2 and Q3 packs have the same table. The Q4 and Q5 packs have a different Table 1, so do not use them here. For question 1, also read the participant flow at the end of section 2. Answer questions 1 to 5 alone, ending with your two Results sentences, and be ready to share them when the instructor asks. Question 6 is for after class.

1. How many patients are in each column? Why is the overall number 618 and not 620?
2. For **three** variables, write the summary used (mean (SD), median (IQR) or n (%)) and why.

Variable	Summary used	Why this summary?

3. Were the two groups alike at the start? Name the **biggest** difference between the negative- and positive-balance groups.
4. Which variable has the most missing values, and how many?
5. Write **two sentences** describing the patients, as they would appear at the start of a Results section (numbers in brackets, no p-values).

6. Looking at this table, would you expect the crude association between positive fluid balance and death to be larger or smaller than the adjusted one? Why?

Exercise 7: [FLUID-ICU] A FLUID-ICU question as a PECO sentence

Task. Fill in the table for Question 1 (Q1), from page 1 of its results pack. Be ready to read your sentence aloud when the instructor asks. At the end of Session 6 you choose one of the five questions: then fill in the table again for that question.

PECO: P = population, E = exposure, C = comparison, O = outcome.

	Your answer
The question (Q1 to Q5), copied from page 1 of its results pack	
P: population	
E: exposure (variable name in the data dictionary) and its type	
C: comparator	
O: outcome (variable name) and its type	
One-sentence research question	
Which test will compare the groups? Which model will adjust? (guess now, check in Session 6)	

Exercise 8: Proposal builder 5 worksheet

Task. Plan the data for your **own** research question, not for FLUID-ICU. First watch the instructor fill in each part for the running example. In class, draft the variables table (8.1, 10 minutes). At home, finish 8.2 to 8.4. Then copy everything into Part 3 of your proposal template (`Research_Proposal_Template.docx`, sent by email).

8.1 Variables table (in class)

Role	Variable	How measured (unit, time point)	Type
Primary outcome			
Secondary outcome (optional)			
Exposure / intervention			
Confounder 1			
Confounder 2			
Confounder 3			
Descriptive (e.g. age, sex)			

Tip: in an RCT, randomisation deals with most confounding. List the key baseline variables you will show in Table 1 instead.

8.2 Mini-CRF: 6–10 items

#	Question exactly as the data collector will read it	Answer format (boxes / ticks / unit)	"Unknown / not done" option?
1			☐
2			☐
3			☐
4			☐
5			☐
6			☐
7			☐
8			☐

8.3 Data dictionary: 3–5 rows and one data-quality plan

Variable name	Label (meaning)	Type	Unit / codes	Allowed values

Data-quality plan (one or two sentences): who checks the forms, how often, and which range or consistency check you will run.

8.4 Data-management plan

Question	Your answer
Paper CRF with double data entry, or electronic data capture (e.g. REDCap / KoBoToolbox)?	
Coded study IDs? Where is the linking log kept, and who can open it?	
Validation rules at entry (two examples from your data dictionary)	
Audit trail / cleaning log: who records changes?	
Backups (3-2-1) and data-sharing statement in the consent form	